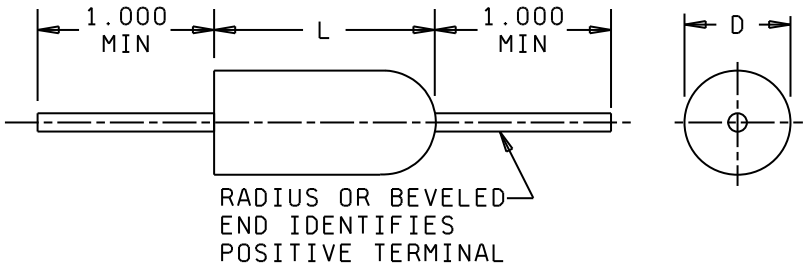
INCH-POUND
MIL-PRF-49137/1E
w/AMENDMENT 1
23 April 2021
SUPERSEDING
MIL-PRF-49137/1E
10 October 2014

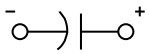
PERFORMANCE SPECIFICATION SHEET

CAPACITOR, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, POLAR, MOLDED, NONHERMETICALLY SEALED, STYLE CX01

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-49137.





CIRCUIT DIAGRAM

Case size	Dimensions		Lead size	
	Diameter D ± .020 (0.51)	Length L ± .020 (0.51)	Dia.	AWG
A	.120 (3.05)	.310 (7.87)	.020 ± .0015 (0.51 ± 0.040)	24
B	.200 (5.08)	.400 (10.16)	.025 ± .002 (0.64 ± 0.05)	22, 24
C	.220 (5.59)	.560 (14.22)	.025 ± .002 (0.64 ± 0.05)	22, 24

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

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TABLE I. Style CX01 characteristics.

Part or identifying number (PIN) 1/	DC rated voltage (+85°C) (Volts)	Capacitance (nominal) (µF)	DC leakage (max) +25°C (µA)	Dissipation Factor (DF) (%)	Case size
CX01D395 -	6	3.9	1.0	6	A
CX01D475 -	6	4.7	1.0	6	A
CX01D565 -	6	5.6	1.0	6	A
CX01D685 -	6	6.8	1.0	6	A
CX01D825 -	6	8.2	1.0	8	A
CX01D106 -	6	10.0	1.0	8	A
CX01D126 -	6	12.0	1.0	8	A
CX01D186 -	6	18.0	1.0	8	B
CX01D686 -	6	68.0	5.0	10	C
CX01F275 -	10	2.7	1.0	6	A
CX01F335 -	10	3.3	1.0	6	A
CX01F226 -	10	22	2.0	8	B
CX01F276 -	10	27	2.0	8	B
CX01F336 -	10	33	2.0	8	B
CX01F396 -	10	39	5.0	8	B
CX01F476 -	10	47	5.0	8	C
CX01F566 -	10	56.0	5.0	8	C
CX01H156 -	15	15	2.0	8	B
CX01H226 -	15	22	3.0	8	C
CX01H276 -	15	27	5.0	8	C
CX01H336 -	15	33	5.0	8	C
CX01J106 -	20	10	1.0	8	B
CX01J126 -	20	12	1.0	8	B
CX01J156 -	20	15	2.0	8	C
CX01J186 -	20	18	5.0	8	C
CX01K155 -	25	1.5	1.0	6	A
CX01K185 -	25	1.8	1.0	6	A
CX01K225 -	25	2.2	1.0	6	A
CX01K565 -	25	5.6	1.5	6	B
CX01K685 -	25	6.8	1.5	6	B
CX01K825 -	25	8.2	2.0	8	B
CX01K126 -	25	12	3.5	8	C
CX01M474 -	35	0.47	1.0	4	A
CX01M564 -	35	0.56	1.0	4	A
CX01M684 -	35	0.68	1.0	4	A
CX01M824 -	35	0.82	1.0	4	A
CX01M105 -	35	1.0	1.0	6	A
CX01M125 -	35	1.2	1.0	6	A
CX01M275 -	35	2.7	1.0	6	B
CX01M335 -	35	3.3	1.0	6	B
CX01M395 -	35	3.9	1.0	6	B
CX01M475 -	35	4.7	1.0	6	B
CX01M685 -	35	6.8	2.0	8	C
CX01M825 -	35	8.2	5.0	8	C
CX01M106 -	35	10.0	5.0	4	C

See footnote at end of table.

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TABLE I. Style CX01 characteristics. Continued.

PIN 1/	DC rated voltage (+85°C) (Volts)	Capacitance (nominal) (μF)	DC leakage (max) +25°C (μA)	Dissipation Factor (DF) (%)	Case size
CX01N104 -	50	0.10	1.0	4	A
CX01N124 -	50	0.12	1.0	4	A
CX01N154 -	50	0.15	1.0	4	A
CX01N184 -	50	0.18	1.0	4	A
CX01N224 -	50	0.22	1.0	4	A
CX01N274 -	50	0.27	1.0	4	A
CX01N334 -	50	0.33	1.0	4	A
CX01N394 -	50	0.39	1.0	4	A
CX01N474 -	50	0.47	1.0	4	B
CX01N564 -	50	0.56	1.0	4	B
CX01N684 -	50	0.68	1.0	4	B
CX01N824 -	50	0.82	1.0	4	B
CX01N105 -	50	1.0	1.0	4	B
CX01N125 -	50	1.2	1.0	6	B
CX01N155 -	50	1.5	1.0	6	B
CX01N185 -	50	1.8	1.0	6	B
CX01N225 -	50	2.2	1.0	6	B
CX01N275 -	50	2.7	1.0	6	C
CX01N335 -	50	3.3	2.0	6	C
CX01N395 -	50	3.9	5.0	6	C
CX01N475 -	50	4.7	5.0	6	C
CX01N565 -	50	5.6	5.0	6	C

1/ Complete PIN will include an additional symbol indicating capacitance tolerance.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case material: Molded, epoxy or plastic.

Leads: Solder - coated metal.

DC rated voltage: See table I.

Operating temperature: -55°C to +85°C.

DC leakage (DCL): See table I.

Capacitance (cap.): See table I.

Cap. tolerance: ±10% (K) or ±20% (M).

Dissipation factor (DF): See table I.

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Stability at low and high temperature: In accordance with MIL-PRF-49137.

Step 1 (+25°C):

DCL: See table I.
ΔCap.: Within tolerance specified in table I.
DF: See table I.

Step 2 (-55°C):

ΔCap: Within ±15 percent of step 1 measured value.
DF: 200 percent of table I.

Step 3 (+25°C):

DCL: See table I.
ΔCap.: Within ±10 percent of step 1 measured value.
DF: See table I.

Step 4 (+85°C):

DCL: 10 times +25°C limit (see table I).
ΔCap.: Within ±15 percent of step 1 measured value.
DF: 150 percent of table I.

Step 5 (+25°C):

DCL: See table I.
ΔCap.: Within ±10 percent of step 1 measured value.
DF: See table I.

Surge voltage: In accordance with MIL-PRF-49137.

DCL: See table I.
ΔCap.: Within ±10 percent of initial value.
DF: See table I.

Life (1,000 hours at +25°C): In accordance with MIL-STD-202-108.

DCL: 200 percent of table I.
ΔCap.: Within ±5 percent of initial measured value.
DF: See table I.

NOTE: These capacitors are intended to be used only where supplemental moisture protection is provided or for noncritical applications where hermetic moisture protection is not required.

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Reference documents. In addition to MIL-PRF-49137, this specification sheet references the following:

MIL-STD-202-108

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5910-2021-010)

Review activities:
Navy - AS, MC, OS, SH
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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